Jacobian Scopes: A Unified Geometric Framework for Token-Level LLM Attribution

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Abstract

Large language models (LLMs) make next-token predictions based on clues present in their context, such as semantic descriptions and in-context examples. Yet, elucidating which prior tokens most strongly influence a given prediction remains challenging due to the proliferation of layers and attention heads in modern architectures. We propose *Jacobian Scopes*, a suite of gradient-based, token-level causal attribution methods for interpreting LLM predictions. Grounded in perturbation theory and information geometry, Jacobian Scopes quantify how input tokens influence various aspects of a model's prediction, such as specific logits, the full predictive distribution, and model uncertainty (effective temperature). Through case studies spanning instruction understanding, translation, and in-context learning (ICL), we demonstrate how Jacobian Scopes reveal implicit political biases, uncover word- and phrase-level translation strategies, and shed light on recently debated mechanisms underlying in-context time-series forecasting. On token-ablation faithfulness benchmarks (LAMBADA and IWSLT2017) across six LLMs, the distribution-level Fisher and Temperature Scopes significantly outperform established gradient-based attribution methods in 10 of 12 model–dataset settings. To facilitate exploration of Jacobian Scopes on custom text, we open-source our implementations and provide a cloud-hosted interactive demo at `https://huggingface.co/spaces/Typony/JacobianScopes`.

1 Introduction

Large Language Models (LLMs) exhibit surprising emergent abilities, such as multi-step [22] and in-context learning (ICL) [10, 15]. However, the mechanistic underpinning of these emergent abilities remains a deep and open question [30, 33, 44]. This has fueled a growing endeavor, across academia and industry, to interpret and explain LLMs' inference process during such emergent thinking modes; as when writing [45, 20] and reasoning [47, 46].

In this work, we present a suite of gradient-based causal attribution methods called *Jacobian Scopes*. Jacobian Scopes explain LLM predictions in terms of influential input tokens. This is achieved by analyzing the Jacobian matrix, which captures the locally linearized relation between the predictive distribution and input embeddings.

Jacobian Scopes come in three variants: **Semantic**, **Fisher**, and **Temperature Scopes**, each corresponding to a different *explanandum*, or attribution objective: logit of a certain token, the full predictive distribution, and the model's uncertainty (effective temperature). An overview of the framework is illustrated in Figure 1.

We showcase the efficacy of our proposed Jacobian Scopes through a range of case studies, including natural language instruction such as system prompts and translation, as well as ICL tasks such as time-series forecasting [18, 29]. We further evaluate attribution faithfulness quantitatively on two

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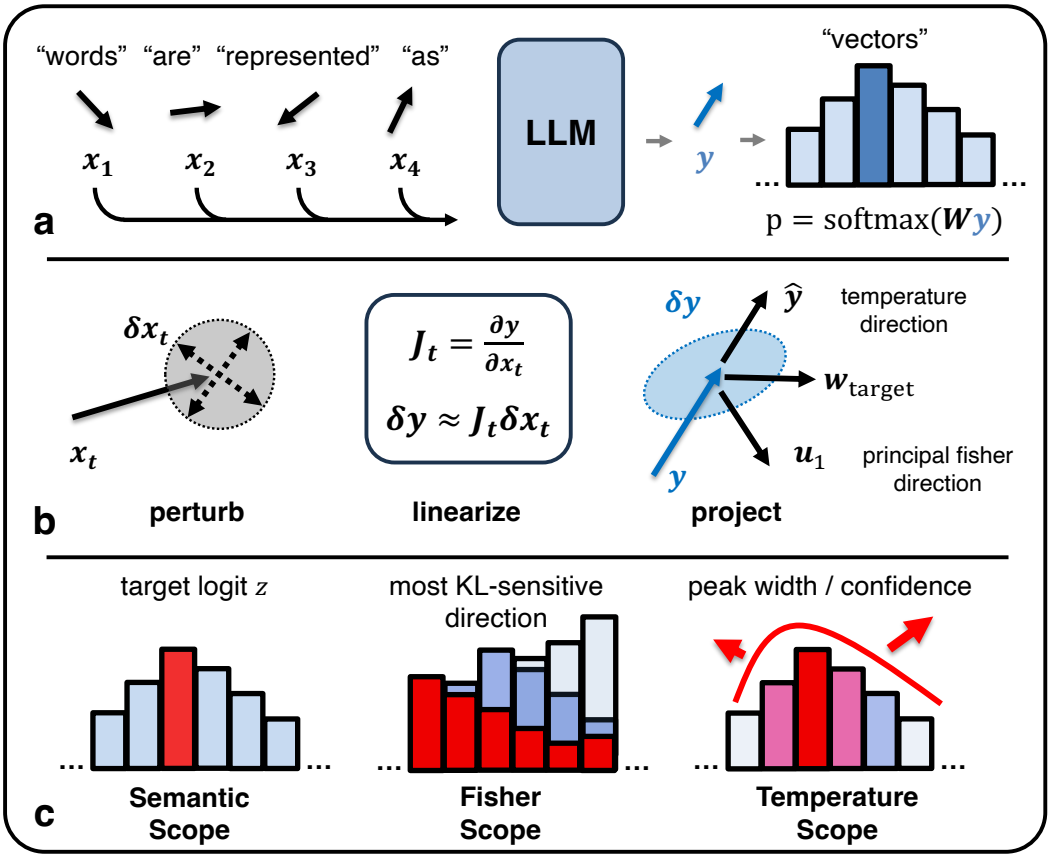


Figure 1: **Overview of Jacobian Scopes. (a)** An autoregressive LLM maps an input token sequence to a hidden state $\boldsymbol{y}$, which is decoded into a predictive distribution over the vocabulary. **(b)** A small perturbation to any input token is propagated to the output via the Jacobian $\boldsymbol{J}_t$, then projected onto a direction $\boldsymbol{v}$ in output space; the magnitude of this projection defines the token's influence score, computable in a single backward pass. The three Scopes correspond to $\boldsymbol{v} = \boldsymbol{w}_{\text{target}}$ (target unembedding row), $\boldsymbol{v} = \boldsymbol{u}_1$ (principal Fisher direction), and $\boldsymbol{v} = \hat{\boldsymbol{y}}$ (normalized hidden state). **(c)** The three Jacobian Scope variants each target a different feature of the model's output: **Semantic Scope** asks which input tokens drive the predicted probability of a specific target word; **Fisher Scope** asks which tokens most change the overall shape of the predicted distribution; and **Temperature Scope** asks which tokens control the model's confidence, i.e., how sharply peaked the distribution is.

benchmark datasets (LAMBADA [31] and IWSLT2017 [7]) across three suites of leading LLMs: LLaMA-3.2 [17], Qwen2.5 [32] and Gemma-3 [42].

2 Related work

Gradient-based attribution methods were first systematically developed for image classifiers. Simonyan et al. [38] introduced class saliency maps by computing the gradient of the predicted class score with respect to the input pixels, identifying which image regions drive a given classification decision in a single backward pass. Building on this, Shrikumar et al. [37] proposed DeepLIFT, which propagates neuron contribution scores by comparing each activation to a reference state; gradient $\times$ input — the elementwise product of the input embedding and its gradient with respect to the target logit — arises as a simpler special case and serves as a baseline in our evaluations. Grad-CAM [36] extends gradient saliency to produce spatially coarser but more semantically coherent maps by pooling gradients across feature channels.

A key challenge with plain gradient saliency is that the resulting maps are often noisy and fail to sharply localize the relevant input features. SmoothGrad [39] addresses this for vision models by averaging sensitivity maps over many copies of the input perturbed with Gaussian noise. Integrated Gradients [41] instead eliminates path-dependence by integrating the gradient along a linear interpolation from a null baseline to the actual input, satisfying axiomatic desiderata such as Sensitivity and Implementation Invariance. Sanyal and Ren [35] subsequently adapted this path integration to the discrete embedding space of language models. Outside the gradient paradigm, SHAP [27] provides model-agnostic attribution scores grounded in Shapley values from cooperative game theory, offering strong axiomatic guarantees but at combinatorial cost that must be approximated in practice.

A complementary line of work interprets transformers through the lens of attention weights. Clark et al. [8] visualize BERT's attention patterns and identify heads that track syntactic and coreference relations. Abnar and Zuidema [1] extend this with attention rollout and attention flow, which propagate attention weights across layers to produce more faithful input-attribution scores. While intuitive and computationally cheap, these attention-based scores do not directly measure the causal influence of input tokens on model outputs, and can be misleading as attribution methods [21]. Ferrando et al. [12] address this for neural machine translation by proposing ALTI (Aggregated LaTent Interaction), a gradient-based method that attributes each generated target token to specific source tokens; our Fisher Scope case study on translation (Figure 3) can be viewed as a distributional generalization of this attribution objective.

More recently, mechanistic interpretability methods have sought to go beyond input-output sensitivity and trace computations through specific model circuits. Activation patching [21, 16] localizes responsible components by intervening on intermediate representations, while circuit tracing [4] reconstructs full computational graphs. Sparse autoencoder (SAE) probes [23] decompose residual stream activations into interpretable features. These methods offer fine-grained mechanistic insights, but are substantially more expensive and architecture-specific than gradient-based approaches.

All gradient-based methods described above attribute a *scalar model output*—a single class logit or token log-likelihood—and are therefore conceptually analogous to *Semantic Scope*. Yet LLMs produce a full distribution over the vocabulary, and no prior methods, to our best knowledge, exploit this richer structure.

3 Methodology

Notations. We use bold lowercase letters (e.g. $\boldsymbol{x}$) for vectors and bold uppercase letters (e.g. $\boldsymbol{W}$) for matrices.

Let $\mathcal{H} := \mathbb{R}^{d_{\text{model}}}$ denote the model's hidden space. We represent a length-T input sequence as $\boldsymbol{X}_{1:T} := (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_t, \ldots, \boldsymbol{x}_T)$, where $\boldsymbol{x}_t \in \mathcal{H}$. We view an auto-regressive LLM as a function that maps a sequence of inputs to the final-layer, post-layer-norm hidden state at the leading position:

$$\boldsymbol{y} := f(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_t, \ldots, \boldsymbol{x}_T) \in \mathcal{H}.$$

The corresponding logits $\boldsymbol{z}$ and predictive distribution $\boldsymbol{p}$ are given by $\boldsymbol{z} := \boldsymbol{W}\boldsymbol{y}$ and $\boldsymbol{p}(\cdot \mid \boldsymbol{X}_{1:T}) := \mathrm{softmax}(\boldsymbol{z}) \in \mathbb{R}^{|\mathcal{V}|}$, where $\boldsymbol{W} \in \mathbb{R}^{|\mathcal{V}| \times d_{\text{model}}}$ denotes the unembedding matrix, and $\mathcal{V}$ is the vocabulary set of size $|\mathcal{V}|$.

Definition of influence score. We are interested in quantifying how perturbations to an input token embedding $\boldsymbol{x}_t$ affect the leading hidden state $\boldsymbol{y}$. We define the input-to-output Jacobian at position t as

$$\boldsymbol{J}_t := \frac{\partial \boldsymbol{y}}{\partial \boldsymbol{x}_t} \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}.$$

Forming the full Jacobian $\boldsymbol{J}_t$ requires d_{model} backward passes, which can be expensive for modern LLMs. Fortunately, in practice we find it sufficient to compute the vector-Jacobian product, $\boldsymbol{v}^\intercal \boldsymbol{J}_t$, where $\boldsymbol{v} \in \mathcal{H}$ captures a property of interest in the predictive distribution, such as confidence (effective inverse temperature) or the embedding of a target token.

Geometrically, $\|\boldsymbol{v}^\intercal \boldsymbol{J}_t\|_2$ measures the largest displacement along the direction $\boldsymbol{v}$, that could be induced by an ε-norm perturbation to $\boldsymbol{x}_t$ (proven in Appendix A.9). We thus define the *influence* of token $\boldsymbol{x}_t$ on the leading output $\boldsymbol{y}$ as

$$\text{Influence}_t := \|\boldsymbol{v}^\intercal \boldsymbol{J}_t\|_2 . \tag{1}$$

Influence computation via auto-diff. For the leading prediction $\boldsymbol{y}$, the influence scores at all input positions $t \leq T$ can be computed with a single backward pass by auto-differentiating the customized scalar loss $\mathcal{L}(\boldsymbol{X}_{1:T}) := \boldsymbol{v}^\intercal \boldsymbol{y}$:

$$\left\| \frac{\partial \mathcal{L}(\boldsymbol{X}_{1:T})}{\partial \boldsymbol{x}_t} \right\|_2 = \left\| \boldsymbol{v}^\intercal \frac{\partial \boldsymbol{y}}{\partial \boldsymbol{x}_t} \right\|_2 = \text{Influence}_t. \tag{2}$$

In section 4, we present three specific choices for the direction of interest $\boldsymbol{v}$, leading to *Semantic Scope*, *Fisher Scope* and *Temperature Scope*; see Figure 1 for an overview.

4 Example applications

This section discusses three formulations of Jacobian Scopes, each motivated by a specific attribution objective. For consistency, all demonstrations in this section are conducted on LLaMA-3.2 1B.

4.1 Semantic Scope

Semantic Scope traces the input tokens that most prominently contribute to the predicted probability of a target vocabulary. Here, the direction of interest is $\boldsymbol{v} = \boldsymbol{w}_{\text{target}}$, which is the row of the unembedding matrix $\boldsymbol{W}$ corresponding to the target token. The customized scalar loss becomes

$$\mathcal{L}_{\text{semantic}} := z_{\text{target}} = \boldsymbol{w}_{\text{target}}^\intercal \boldsymbol{y}, \tag{3}$$

which is exactly the predicted logit of the target token. Differentiating this loss using eq. (2) yields the Semantic Scope influence score

$$\text{Influence}_t^{\text{Sem}} := \left\| \boldsymbol{w}_{\text{target}}^\intercal \boldsymbol{J}_t \right\|_2. \tag{4}$$

Intuitively, Semantic Scope quantifies how sensitive the target logit is to changes in specific input tokens. Figure 2 demonstrates how Semantic Scope may be employed to visualize implicit political bias in LLMs (see also Figure 6 in the appendix for a system-prompt example).

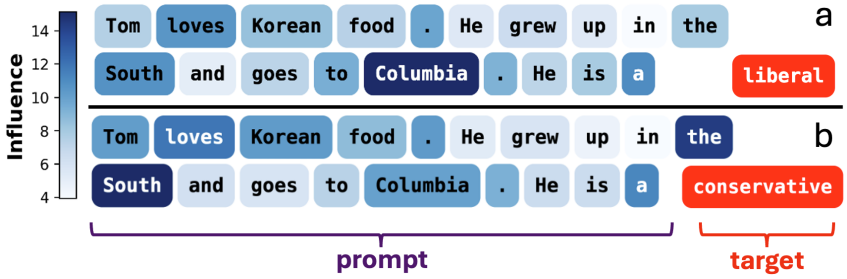


Figure 2: Semantic Scope reveals the potential for political bias that may be present in models such as LLaMA-3.2. The prediction for the subject being a "liberal" is attributed to the input token "Columbia", while "conservative" to the token pairs for "the South".

4.2 Fisher Scope

Semantic Scope attributes the past tokens that affect a specific output logit, but it does not capture their influence on the *entire* predictive distribution $\boldsymbol{p}(\cdot \mid \boldsymbol{X}_{1:T})$. The latter motivates Fisher Scope, which is particularly suited for attribution tasks where the correct next-word prediction is non-unique, such as translation (see Figure 3(b)).

Fisher Scope is inspired by information geometry [13], which studies how infinitesimal changes in model parameters affect the predictive distributions [6]. In our setting, the hidden state $\boldsymbol{y}$ parameterizes the family $\boldsymbol{p}(\boldsymbol{y}) = \text{softmax}(\boldsymbol{W}\boldsymbol{y})$, and the Fisher information matrix (FIM) $\boldsymbol{F}$, derived in appendix A.10, measures how sensitively $\boldsymbol{p}$ responds to perturbations in $\boldsymbol{y}$:

$$\boldsymbol{F} := \boldsymbol{W}^\intercal \big(\text{diag}(\boldsymbol{p}) - \boldsymbol{p}\boldsymbol{p}^\intercal\big)\boldsymbol{W}. \tag{5}$$

Intuitively, the spectrum of $\boldsymbol{F}$ reveals how sensitively a perturbation along each eigendirection can change the predictive distribution. Formally, the eigen-decomposition $\boldsymbol{F} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^\intercal$ assigns to each direction $\boldsymbol{u}_i$ a scalar sensitivity λ_i, quantifying how strongly a perturbation along $\boldsymbol{u}_i$ changes $\boldsymbol{p}(\cdot \mid \boldsymbol{X}_{1:T})$, as measured by KL divergence [3].

We call the leading eigenvector $\boldsymbol{u}_1$ the *principal Fisher direction*: it is the single direction in output space $\mathcal{V}$ along which a perturbation to $\boldsymbol{y}$ most sensitively changes $\boldsymbol{p}(\cdot \mid \boldsymbol{X}_{1:T})$. The Fisher Scope influence score is then defined as the projection of the Jacobian onto this direction:

$$\text{Influence}_t^{\text{Fisher}} := \|\boldsymbol{u}_1^\top \boldsymbol{J}_t\|_2. \tag{6}$$

In appendix A.11, we show that Fisher Scope can be viewed as a rank-1 approximation of the total information between predictive distribution $\boldsymbol{p}$ and input embedding $\boldsymbol{x}_t$.

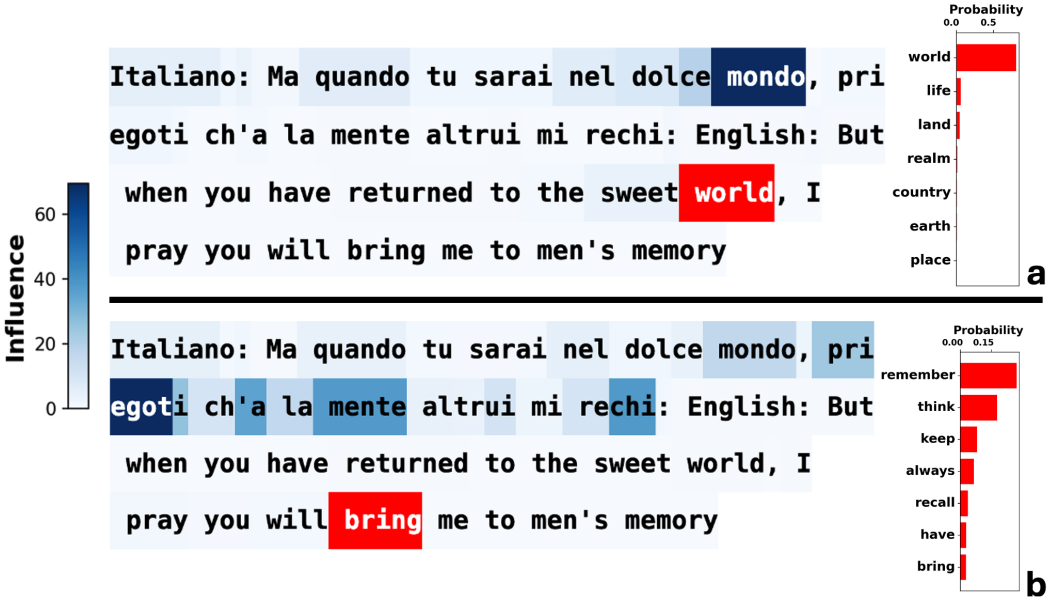


Figure 3: Fisher Scope visualizes LLaMA-3.2 3B's strategy for translating Dante's Inferno [2]. Left: Magnitude of influence from each source word (blue) on the predicted distribution at each target position (red). Right: Probability distribution of top 7 words predicted at each target position. (a) LLaMA assigns high probability to "world", the unambiguous literal translation of "mondo". (b) When translating the phrase "priegoti ch'a la mente altrui mi rechi" (literally: "I pray thee to bring me to another's mind"), LLaMA's predicted distribution is highly uncertain, with each top token hinting at a different phrasing strategy.

Figure 3 demonstrates how Fisher Scope reveals two non-exclusive translation strategies employed by LLaMA-3.2. (a) When the source text has precise word-level counterparts, the LLM opts for word-to-word translation, with each predicted token attributed to its lexical counterpart. (b) When word-to-word translation is not possible due to mismatching syntactical structures, the LLM works at the level of phrases, with each predicted position attending to multiple source tokens. Fisher Scope is particularly well-suited to case (b): because the predicted distribution becomes highly uncertain across many plausible phrasings, a target-logit method like Semantic Scope would only attribute the influence on one arbitrarily chosen token, whereas Fisher Scope captures the full distributional uncertainty and attributes it to the relevant source phrase. We refer the reader to appendix A.4 for a simpler word-level translation example.

4.3 Temperature Scope

While semantic next-word predictions span a large and heterogeneous vocabulary, structured ICL tasks, such as time-series prediction, [18] often exhibit structural regularities: the predicted distributions form near-Gaussian peaks over a small subset of numerical tokens (see Figures 4 and 13 as well

as [25]). Temperature Scope identifies which input tokens control the width of this Gaussian peak, and hence the model's forecasting uncertainty.

To define Temperature Scope, we first decompose the hidden state $\boldsymbol{y}$ into its norm and direction, $\boldsymbol{y} = \|\boldsymbol{y}\|_2 \hat{\boldsymbol{y}}$, so that the logit vector could be written as

$$\boldsymbol{z} = \boldsymbol{W}\boldsymbol{y} = \|\boldsymbol{y}\|_2 \, (\boldsymbol{W}\hat{\boldsymbol{y}}) = \beta_{\text{eff}} \, \hat{\boldsymbol{z}},$$

where $\beta_{\text{eff}} := \|\boldsymbol{y}\|_2$ is the effective inverse temperature and $\hat{\boldsymbol{z}}$ the normalized logits.

The key theoretical justification is that β_{eff} directly controls predictive spread. Treating token values v as continuous, let $p(v) \propto e^{\,\beta_{\text{eff}}\,\hat{z}(v)}$ be the predicted distribution.

Proposition 1 (Effective temperature as inverse variance)**.** *$p(v)$ is approximately Gaussian if and only if $\hat{z}(v)$ is quadratic in the neighborhood of the peak. Writing $\hat{z}(v) = -b(v-\mu)^2 + c$ with $b > 0$, the predictive variance satisfies*

$$\sigma^2 = \frac{1}{2\,\beta_{\text{eff}}\,b}, \tag{7}$$

so $\beta_{\text{eff}} \propto \sigma^{-2}$, with b a model-dependent constant.

The proof follows directly from matching the Gaussian kernel to the Boltzmann form; see appendix A.7. Proposition 1 is intended for the numerical ICL setting: when the predictive distribution over numerical tokens is approximately Gaussian (as in Figure 4), attributing β_{eff} is equivalent to attributing predictive variance. This motivates using $\mathcal{L}_{\text{temperature}} = \beta_{\text{eff}}$ as the customized loss, whose auto-differentiation yields the influence score:

$$\text{Influence}_t^{\text{Temp}} := \|\hat{\boldsymbol{y}}^{\intercal}\boldsymbol{J}_t\|_2 \,. \tag{8}$$

Entropy interpretation for general vocabularies. For unstructured natural-language prediction, a continuous-token approximation is undefined, so proposition 1 does not apply directly. Temperature Scope is nonetheless competitive on LAMBADA and IWSLT2017 (table 1). A simple explanation is that, for any discrete softmax with fixed logit *shape*, attributing inverse temperature is equivalent to attributing Shannon entropy—and hence predictive uncertainty.

Write $p(i;\beta) \propto e^{\beta\hat{z}(i)}$ with normalized logits $\hat{\boldsymbol{z}}$ held fixed, and let $V(\beta) = \text{Var}_{i\sim p(\cdot;\beta)}[\hat{z}(i)]$. Differentiating the Shannon entropy $H(\beta) = -\sum_i p(i;\beta)\log p(i;\beta)$ gives

$$\frac{dH}{d\beta} = -\beta\, V(\beta). \tag{9}$$

Whenever $\beta > 0$ and $\hat{\boldsymbol{z}}$ is non-constant, $V(\beta) > 0$, so H is strictly decreasing in β: $H \to \log|\mathcal{V}|$ as $\beta \to 0$ and $H \to 0$ as $\beta \to \infty$. Thus $\beta \mapsto H$ is invertible, and attributing β_{eff} is equivalent to attributing H.

The correspondence extends to token-level scores. Temperature Scope computes the radial sensitivity of β_{eff} to each input while holding $\hat{\boldsymbol{z}}$ fixed, so

$$\frac{\partial H}{\partial \boldsymbol{x}_t} = -\beta\, V(\beta)\, \frac{\partial \beta_{\text{eff}}}{\partial \boldsymbol{x}_t} \qquad (\hat{\boldsymbol{z}} \text{ fixed}). \tag{10}$$

The prefactor $-\beta V(\beta)$ is shared by every token, so Temperature Scope and entropy attribution induce the same gradient-norm ranking. In short, there are two complementary readings of the same radial Jacobian projection: on numerical ICL, Temperature Scope attributes predictive *variance*; on general text, it attributes predictive *entropy*.

To probe how an LLM reasons over its inputs during numerical ICL tasks, we follow Gruver et al. [18] and tokenize time-series data as alternating commas and 2-digit numbers,[1] yielding purely numerical prompts such as $29, 30, 45, \ldots$ Temperature Scope then attributes the confidence of the predicted next-state distribution to the inputs.

As shown in Figure 4, when extrapolating chaotic time-series data with recurring but non-identical motifs, LLaMA exhibits consistent attribution patterns: it attends to segments in history that match the local pattern near the cutoff point. This attribution signature directly supports the recent conjecture

[1] In LLaMA-3.2, there is one token for each 2-digit number from 10 to 99.

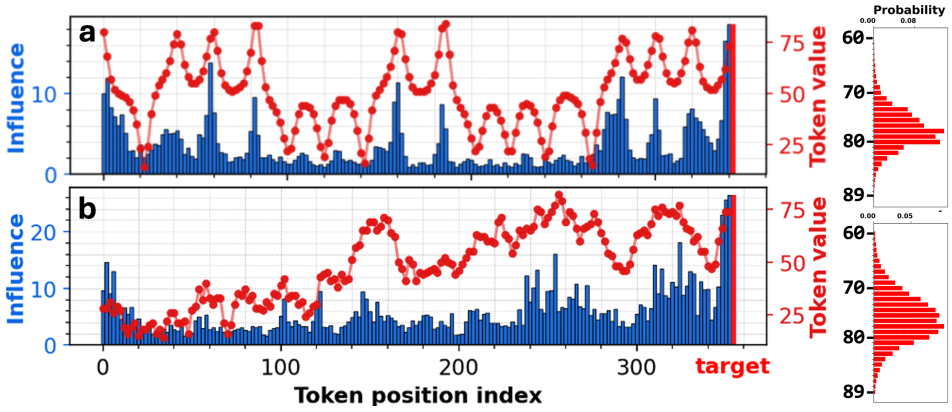


Figure 4: Left: Temperature Scope reveals different strategies employed by LLaMA-3.2 during time-series forecasting: (a) when continuing the dynamics of a partially observed Lorenz system, LLaMA attends to regions in the input history that exhibit patterns similar to those near the cutoff. (b) When extrapolating unstructured Brownian motion, LLaMA shows diminished influence of early context. Right: Predicted probability distribution over the top 30 most probable tokens.

that LLMs might parrot the best-matching pattern in history via nearest-neighbor search in delayed embedding space [49].

On the other hand, when faced with stochastic systems with no recurring motifs, such as Brownian motion, the LLM employs a different strategy: it preferentially attends to later parts of the context. This strategy explains the observation [24] that the ICL loss curve plateaus early for unbounded stochastic systems such as Brownian motion and Geometric Brownian Motion: as such systems wander out of distribution, the LLM ceases to recognize earlier history as relevant, and gradually forgets earlier statistics. We investigate this hypothesis further in appendix A.5.

Appendix A.6 further shows that Temperature Scope produces more interpretable attributions of in-context forecasting behaviour than Semantic and Fisher Scopes.

5 Quantitative Evaluations

To complement the qualitative case studies in section 4, we provide quantitative evaluations against established attribution baselines across two benchmarks.

Datasets. We evaluate on two tasks that probe complementary aspects of token-level attribution.

- **LAMBADA** [31] is a language modeling benchmark designed to test long-range context comprehension. The dataset is constructed such that a human subject would need to locate clues scattered across many sentences in order to correctly predict the final word of each passage.
- **IWSLT2017 DE→EN** [7] is a machine translation dataset that contains passages in German followed by English translations.

Together, these datasets test whether Jacobian Scopes produce faithful attributions across both monolingual and cross-lingual generation. We compare against random ablation, Integrated Gradients [41], and Input $\times$ Gradient [37, 38]. Full experimental details are provided in appendix A.1; PAD-token ablation and the full paired-test table appear in appendices A.2 and A.3. Snapshots of each dataset and their respective Fisher Scope attributions are shown in Figure 5 in the appendix, giving a sense of the typical prompt structure and attribution patterns.

Evaluation Metric. We evaluate using the **Area Over the Perturbation Curve (AOPC)** [34]. AOPC is a counterfactual, interventional test: the top-$k\%$ most influential tokens are ablated (replaced with zero vectors), and the resulting drop in the ground-truth target token's log-probability is recorded.

If an attribution method correctly identifies the tokens that the model relies on, removing them should cause a large, predictable drop in model confidence. This makes AOPC a particularly demanding

benchmark for gradient-based methods such as Jacobian Scopes, which compute only a local, first-order approximation to token influence: performing well on this interventional metric implies that first-order sensitivity is a reliable proxy for genuine causal relevance.

For robustness and consistency with literature [34, 9] we zero out the most influential tokens at three distinct ablation rates, $k \in \{5\%, 10\%, 20\%\}$. The final AOPC is then aggregated via the trapezoidal rule:

$$\mathrm{AOPC} = \sum_i \tfrac{1}{2}(k_{i+1} - k_i)\left(\Delta \log p_{k_i} + \Delta \log p_{k_{i+1}}\right), \tag{11}$$

where $\Delta \log p_k$ is the drop in log-probability when the top-$k\%$ tokens are ablated. A more negative AOPC indicates higher attribution faithfulness.

Results. Table 1 reports AOPC across all six models at both 3B/4B and 1B/1.5B scales. Jacobian Scopes outperform random ablation and match or exceed the strongest baseline in most settings; the main exception is Gemma-3 4B on IWSLT2017, where Integrated Gradients (20-point path integral) attains the best mean AOPC. To assess whether these gaps are reliable at the passage level, we compare, for each model–dataset cell, the mean over passages of $\mathrm{AOPC}_i(\mathrm{Scope}) - \mathrm{AOPC}_i(\mathrm{baseline})$, where the Scope and non-Scope baseline are the best methods in that cell of table 1. Negative values favor the Scope. The Scope wins significantly in **10 of 12** combinations, with one inconclusive comparison and one significant baseline win (Gemma-3 4B IWSLT2017); see table 4 for method pairs, confidence intervals, and p-values. As a robustness check, we also rerun AOPC with length-preserving PAD-token ablation in appendix A.2. We discuss this model- and task-specific pattern in section 6.

Table 1: AOPC (higher magnitude is better) on LAMBADA (LMB) and IWSLT2017 DE→EN (IWS) for all six models, averaged over ablation fractions $k \in \{5\%, 10\%, 20\%\}$ via the trapezoidal rule over 1000 test passages.

1B/1.5B-scale

	LLaMA-3.2 1B		**Qwen2.5-1.5B**		**Gemma-3 1B**	
Method	**LMB**	**IWS**	**LMB**	**IWS**	**LMB**	**IWS**
Random	-0.26 ± 0.01	-0.30 ± 0.01	-0.35 ± 0.01	-0.25 ± 0.01	-0.42 ± 0.01	-0.41 ± 0.01
Integrated Grads	-1.19 ± 0.01	-1.04 ± 0.01	-1.71 ± 0.01	-1.36 ± 0.01	-0.69 ± 0.01	-0.87 ± 0.01
Input × Grad	-1.28 ± 0.01	-1.04 ± 0.01	-1.72 ± 0.01	-1.35 ± 0.01	-1.56 ± 0.01	-1.36 ± 0.01
Semantic Scope	-1.30 ± 0.01	-1.06 ± 0.01	-1.68 ± 0.01	-1.24 ± 0.01	-1.64 ± 0.01	-1.30 ± 0.01
Temperature Scope	$\mathbf{-1.32 \pm 0.01}$	$\mathbf{-1.09 \pm 0.01}$	$\mathbf{-1.86 \pm 0.01}$	$\mathbf{-1.38 \pm 0.01}$	-1.66 ± 0.01	-1.29 ± 0.01
Fisher Scope	$\mathbf{-1.32 \pm 0.01}$	-1.08 ± 0.01	-1.77 ± 0.01	$\mathbf{-1.38 \pm 0.01}$	$\mathbf{-1.67 \pm 0.01}$	$\mathbf{-1.39 \pm 0.01}$

3B/4B-scale

	LLaMA-3.2 3B		**Qwen2.5 3B**		**Gemma-3 4B**	
Method	**LMB**	**IWS**	**LMB**	**IWS**	**LMB**	**IWS**
Random	-0.23 ± 0.01	-0.19 ± 0.01	-0.29 ± 0.01	-0.20 ± 0.01	-0.29 ± 0.01	-0.18 ± 0.01
Integrated Grads	-0.69 ± 0.01	-0.61 ± 0.01	-1.39 ± 0.01	-1.12 ± 0.01	-1.50 ± 0.02	$\mathbf{-1.34 \pm 0.01}$
Input × Grad	-1.12 ± 0.01	-0.77 ± 0.01	-1.39 ± 0.01	-1.11 ± 0.01	-1.70 ± 0.02	-1.17 ± 0.01
Semantic Scope	-1.16 ± 0.01	-0.78 ± 0.01	-1.34 ± 0.01	-1.01 ± 0.02	-1.78 ± 0.02	-1.06 ± 0.01
Temperature Scope	$\mathbf{-1.17 \pm 0.01}$	-0.76 ± 0.01	$\mathbf{-1.55 \pm 0.01}$	$\mathbf{-1.20 \pm 0.01}$	-1.78 ± 0.02	-0.99 ± 0.01
Fisher Scope	$\mathbf{-1.17 \pm 0.01}$	$\mathbf{-0.80 \pm 0.01}$	-1.41 ± 0.01	-1.15 ± 0.01	$\mathbf{-1.81 \pm 0.01}$	-1.20 ± 0.01

These results reveal a perhaps surprising finding: Fisher and Temperature Scopes, which are entirely *target-blind*, match or outperform the target-specific methods (Semantic Scope, Input × Gradient, Integrated Gradients) under AOPC in most settings, despite AOPC being an intrinsically target-specific metric that measures the drop in the ground-truth token's log-probability upon ablation.

One might expect target-specific methods to have an inherent advantage here. The fact that they do not suggests a deeper principle:

> The input tokens that dominate the full predictive distribution are also the tokens that most strongly influence any high-likelihood prediction within that distribution.

In other words, distributional influence is a more reliable proxy for causal relevance than logit-wise sensitivity. This has a practical implication: when the correct next token is unknown or non-unique, as in open-ended generation or ambiguous translation, Fisher and Temperature Scopes provide faithful attributions without requiring a target to be specified.

Table 2: Comparison of the Jacobian Scopes methods introduced. $\boldsymbol{w}_{\text{target}}$ denotes the target vocab embedding, $\boldsymbol{u}_1$ the principal Fisher direction, and $\hat{\boldsymbol{y}}$ the normalized hidden direction.

	Semantic Scope	**Fisher Scope**	**Temperature Scope**
Quantity explained	target logit z	full distribution $\boldsymbol{p}$	confidence $\|\boldsymbol{y}\|_2$
Influence score	$\|\boldsymbol{w}_{\text{target}}^{\intercal}\boldsymbol{J}_t\|_2$	$\|\boldsymbol{u}_1^{\intercal}\boldsymbol{J}_t\|_2$	$\|\hat{\boldsymbol{y}}^{\intercal}\boldsymbol{J}_t\|_2$
Recommended use case	semantic reasoning & bias analysis	NLP tasks with non-unique predictions	in-context learning on numerical data

The primary strength of Jacobian Scopes therefore resides not only in its geometric motivations and single-backward-pass efficiency, but also in the native distributional awareness that proves empirically superior in terms of attribution faithfulness.

6 Discussion and conclusion

As summarized in table 2, the three Jacobian Scopes span a spectrum from logit-specific (**Semantic**) to fully distribution-aware (**Fisher**, **Temperature**), each targeting a distinct feature of the model's output: the logit of a specific token, the shape of the full predictive distribution, and the sharpness of that distribution, respectively. This spectrum directly addresses the gap identified in section 2: while all prior gradient-based methods attribute a scalar output and are therefore analogous to Semantic Scope, Fisher and Temperature Scopes are the first, to our knowledge, to natively attribute the full predictive distribution. *Fisher Scope* does so by projecting onto the principal direction of the Fisher information matrix, which is the single hidden-space perturbation that most changes the distribution under KL divergence. *Temperature Scope*, on the other hand, attributes the effective inverse temperature: predictive variance in numerical ICL (proposition 1) and, equivalently, predictive entropy for general softmax distributions (eqs. (9) and (10)).

Quantitative evaluations confirm that this distributional awareness leads to empirical gains: across both LAMBADA and IWSLT2017, Fisher and Temperature Scopes significantly outperform the strongest baseline in 10 of 12 model–dataset cells (table 4). The exception on Gemma-3 4B IWSLT2017, where the 20-point Integrated Gradients baseline wins the paired comparison, points to an open question: the relative faithfulness of different attribution methods may be model-architecture-specific. Gemma-3's 5:1 interleaved local/global attention [42] creates qualitatively different gradient pathways for tokens at different context offsets, which may interact differently with first-order versus path-integrated attribution strategies, particularly in cross-lingual settings where source tokens are systematically offset from the target. Investigating how architectural choices, such as sliding-window attention, grouped-query attention, or non-standard positional encodings may modulate the reliability of gradient-based attribution methods could yield new insights.

More broadly, we put forward vector-Jacobian projection as a unifying framework for gradient-based attribution in LLMs. Any choice of projection vector $\boldsymbol{v}$ defines a new explanandum, and the three Scopes presented here are particular instances of this family. A profitable direction for future work is to derive new members by studying the spectral structure of the Jacobian $\boldsymbol{J}_t$ and the Fisher information matrix $\boldsymbol{F}$, potentially yielding attribution methods tailored to other aspects of LLM behaviours such as calibration [19], hallucination [43], or factual recall [28].

Limitations

Linearized causality. Jacobian Scopes extract first-order input-output relationships via automatic differentiation, yielding local causal attributions in the linear neighborhood of the input. As such, they are closer in spirit to gradient-based methods such as SmoothGrad [39] and Integrated Gradients [41, 35] than to explicitly interventional approaches like activation patching [21, 16] or circuit tracing [4].

Architecture-blindness. Jacobian Scopes characterize input–output sensitivity without reference to a model's internal transformer architecture. This makes the analysis more parsimonious, but also restricts the mechanistic insights [8]. As a result, Jacobian Scope attributions should be interpreted with awareness of potential circuit-level eccentricities at play [4, 23]. For instance, Temperature Scope attributions in Figure 4 (b) show elevated influence scores for early context tokens, an effect we attribute to the attention sink phenomenon [48]. Appendix A.8 discusses further complications due to attention sink.

Dependence on back-propagation. Like most gradient-based attribution methods, Jacobian Scopes rely on back-propagation, which is in theory more computationally intensive than forward-only approaches such as those based on pretrained auto-encoders [23] or attention visualization [8].

In practice, however, the overhead is negligible: on a single NVIDIA RTX A4000, the backward pass for the example in Figure 7 takes only 0.027 seconds, compared to 0.069 seconds for the forward pass. This efficiency follows directly from the design of Jacobian Scopes: rather than backpropagating through all model parameters, each attribution requires only a single vector-Jacobian product with respect to the input embeddings, with parameter gradients disabled throughout.

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A Appendix

A.1 Additional information on Benchmarks

Datasets and preprocessing. LAMBADA [31] is a dataset of narrative passages specifically designed so that a human reader would need to process the *full* context in order to predict the final word — local context alone is insufficient. This property makes it ideal for evaluating token-level attribution methods, as a faithful attribution should identify long-range contextual dependencies rather than trivial local patterns. We retain only passages of at least 50 words, and remove any trailing punctuation from the final word.

IWSLT2017 DE→EN [7] consists of German-to-English translation pairs from TED talks. Each example is formatted as `German: [source]. English: [translation]`, and we base the prediction on the final non-punctuation token of the English translation. Faithful attribution requires long-range attention: the model must attend to the corresponding German source tokens across the full prompt. We retain only examples whose English translation contains at least 20 words. We evaluate on 1000 passages per dataset.

Evaluation metric. We evaluate using the Area Over the Perturbation Curve (AOPC) [34]. AOPC measures attribution faithfulness by progressively ablating the most influential tokens as deemed by a certain attribution method, and recording the resulting drop in the ground-truth target token's log-probability. Ablated tokens are replaced with zero vectors. Formally, for ablation fractions $k \in \{5\%, 10\%, 20\%\}$, the AOPC is computed using the trapezoidal rule

$$\mathrm{AOPC} = \sum_i \frac{1}{2}(k_{i+1} - k_i)\left(\Delta \log p_{k_i} + \Delta \log p_{k_{i+1}}\right),$$

where $\Delta \log p_k$ denotes the drop in log-probability when the top-$k\%$ most influential tokens are ablated. A more negative AOPC indicates that the ablated tokens were genuinely influential, reflecting higher attribution quality. The $\pm$ values in all tables report the standard error of the mean across the 1000 test passages.

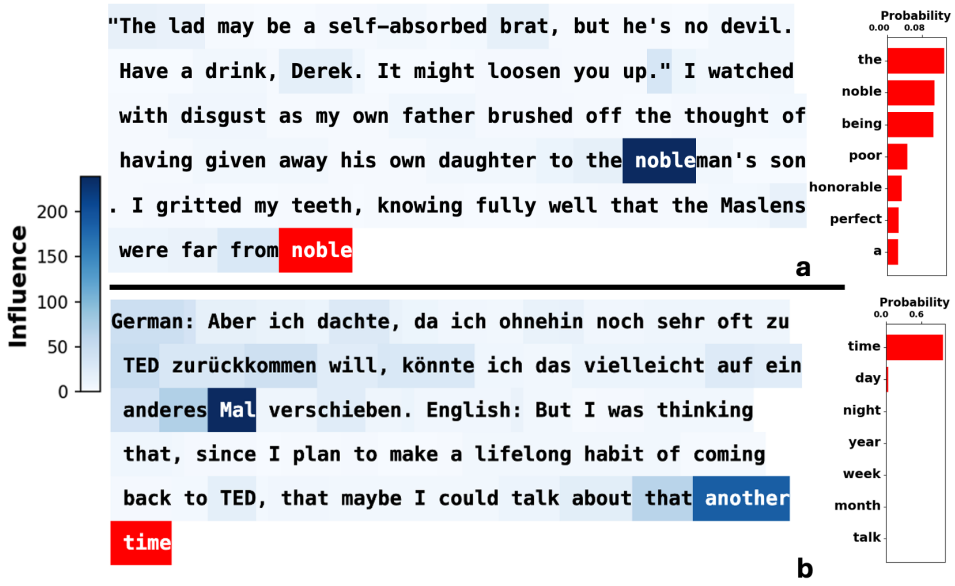


Figure 5: Fisher Scope attribution examples on (a) LAMBADA and (b) IWSLT2017 DE→EN, using LLaMA-3.2 3B. Darker blue indicates higher influence; the target token is highlighted in red. Right: Probability distribution of top 7 words predicted at target position.

Baselines. *Input* × *Gradient* [37] defines the influence score of token $\boldsymbol{x}_t$ as the ℓ_2 norm of the element-wise product of the input embedding and its gradient with respect to the target logit:

$$\text{Influence}_t^{\text{I}\times\text{G}} := \|\boldsymbol{x}_t \odot \nabla_{\boldsymbol{x}_t} z_{\text{target}}\|_2 \,. \tag{12}$$

Integrated Gradients [41], defined in eq. (15), integrates the gradient along a linear interpolation path from null (all zero) input to the full input. In the AOPC benchmark of table 1, we approximate the path integral with a 20-point right-endpoint grid $\alpha \in \{0.05, 0.10, \ldots, 1.00\}$. We refer the reader to appendix A.8, which visualizes the path integration in detail and discusses why this approach introduces distortions in the LLM setting — most notably attention sink effects at small interpolation values — that likely contribute to its weaker AOPC performance.

Jacobian Scopes. We evaluate all three Jacobian Scope variants introduced in section 4.

Similar to Input × Gradient and Integrated Gradients, *Semantic Scope* is target-specific: the direction of interest is $\boldsymbol{v} = \boldsymbol{w}_{\text{target}}$, the unembedding row corresponding to the ground-truth next token, giving

$$\begin{aligned} \text{Influence}_t^{\text{Sem}} &:= \left\|\boldsymbol{w}_{\text{target}}^{\intercal} \boldsymbol{J}_t\right\|_2 \\ &= \|\nabla_{\boldsymbol{x}_t} z_{\text{target}}\|_2 \,. \end{aligned} \tag{13}$$

Fisher Scope and *Temperature Scope*, by contrast, are distribution-aware: they attribute to the entire predictive distribution $\boldsymbol{p}(\cdot \mid \boldsymbol{X}_{1:T})$ rather than a single logit, as defined in sections 4.2 and 4.3.

A.2 PAD-token ablation

The main AOPC evaluation in table 1 replaces ablated token embeddings with zeros. To test sensitivity to this intervention, we rerun the full benchmark with a length-preserving PAD-token replacement: selected embeddings are replaced by a model-family padding embedding while attention-mask entries remain active, so sequence length and positional structure are preserved. Filler tokens are LLaMA <|finetune_right_pad_id|>, Qwen's shared PAD/EOS <|endoftext|>, and Gemma <pad>. We avoid BOS/EOS fillers because they can act as attention sinks [48].

Table 3 reports the resulting AOPC. A Jacobian Scope attains the best or tied-best mean in all 12 model–dataset cells, and on LLaMA and Qwen the ranking relative to baselines is preserved. Thus the central Scope-versus-baseline conclusion is robust to replacing zero ablation with PAD-token ablation.

Table 3: AOPC under PAD-token ablation (higher magnitude is better) on LAMBADA (LMB) and IWSLT2017 DE→EN (IWS), averaged over $k \in \{5\%, 10\%, 20\%\}$ via the trapezoidal rule. Best or tied-best entries in each column are bolded.

	1B/1.5B-scale					
	LLaMA-3.2 1B		**Qwen2.5-1.5B**		**Gemma-3 1B**	
Method	**LMB**	**IWS**	**LMB**	**IWS**	**LMB**	**IWS**
Random	-0.41 ± 0.01	-0.36 ± 0.01	-0.51 ± 0.02	-0.43 ± 0.02	-0.63 ± 0.01	-0.85 ± 0.01
Integrated Grads	-1.36 ± 0.02	-1.27 ± 0.02	-2.06 ± 0.02	-2.08 ± 0.03	$\mathbf{-0.81 \pm 0.01}$	-1.02 ± 0.01
Input × Grad	-1.61 ± 0.02	-1.46 ± 0.02	-2.06 ± 0.02	-2.09 ± 0.03	-0.80 ± 0.01	-1.03 ± 0.01
Semantic Scope	-1.63 ± 0.02	-1.49 ± 0.02	-2.00 ± 0.02	-1.92 ± 0.03	-0.80 ± 0.01	-1.03 ± 0.02
Temperature Scope	$\mathbf{-1.69 \pm 0.02}$	$\mathbf{-1.53 \pm 0.02}$	$\mathbf{-2.34 \pm 0.02}$	$\mathbf{-2.20 \pm 0.03}$	$\mathbf{-0.81 \pm 0.01}$	$\mathbf{-1.04 \pm 0.02}$
Fisher Scope	$\mathbf{-1.69 \pm 0.02}$	$\mathbf{-1.53 \pm 0.02}$	-2.11 ± 0.02	-2.11 ± 0.03	-0.80 ± 0.01	-1.03 ± 0.02
	3B/4B-scale					
	LLaMA-3.2 3B		**Qwen2.5 3B**		**Gemma-3 4B**	
Method	**LMB**	**IWS**	**LMB**	**IWS**	**LMB**	**IWS**
Random	-0.34 ± 0.01	-0.29 ± 0.01	-0.54 ± 0.02	-0.45 ± 0.02	-0.25 ± 0.01	-0.14 ± 0.00
Integrated Grads	-0.77 ± 0.02	-0.70 ± 0.02	-1.80 ± 0.02	-1.80 ± 0.03	-0.51 ± 0.01	-0.42 ± 0.01
Input × Grad	-1.32 ± 0.02	-1.02 ± 0.02	-1.81 ± 0.02	-1.80 ± 0.03	-0.51 ± 0.01	-0.43 ± 0.01
Semantic Scope	-1.35 ± 0.02	-1.02 ± 0.02	-1.73 ± 0.02	-1.58 ± 0.03	-0.51 ± 0.01	-0.42 ± 0.01
Temperature Scope	-1.35 ± 0.02	-1.00 ± 0.02	$\mathbf{-2.24 \pm 0.02}$	$\mathbf{-2.02 \pm 0.02}$	$\mathbf{-0.54 \pm 0.01}$	$\mathbf{-0.48 \pm 0.01}$
Fisher Scope	$\mathbf{-1.37 \pm 0.02}$	$\mathbf{-1.05 \pm 0.02}$	-1.86 ± 0.02	-1.85 ± 0.03	-0.52 ± 0.01	-0.46 ± 0.01

A.3 Full paired significance results

Table 4 reports passage-level paired differences between the best Scope and best non-Scope baseline in each cell of table 1. For each passage i we compute $\mathrm{AOPC}_i(\mathrm{Scope}) - \mathrm{AOPC}_i(\mathrm{baseline})$; the table lists the mean difference, a 95% Student-t confidence interval, and a two-sided Wilcoxon p-value. Methods are fixed from table 1 before testing. Negative Δ favors the Scope.

Table 4: Paired AOPC tests (best Scope vs. best baseline from table 1). Negative Δ favors the Scope; bold entries are significant at $p < 0.05$ (two-sided Wilcoxon).

Model	**Dataset**	**Scope**	**Baseline**	**Mean Δ [95% CI]**	p
LLaMA-3.2 1B	LAMBADA	Fisher	Input × Grad	$\mathbf{-0.036}$ $[\mathbf{-0.048}, \mathbf{-0.024}]$	$\mathbf{1.2\times10^{-6}}$
LLaMA-3.2 1B	IWSLT	Temperature	IG	$\mathbf{-0.049}$ $[\mathbf{-0.073}, \mathbf{-0.025}]$	$\mathbf{4.0\times10^{-4}}$
LLaMA-3.2 3B	LAMBADA	Fisher	Input × Grad	$\mathbf{-0.055}$ $[\mathbf{-0.071}, \mathbf{-0.039}]$	$\mathbf{1.1\times10^{-11}}$
LLaMA-3.2 3B	IWSLT	Fisher	Input × Grad	$\mathbf{-0.029}$ $[\mathbf{-0.042}, \mathbf{-0.015}]$	$\mathbf{3.9\times10^{-4}}$
Qwen2.5 1.5B	LAMBADA	Temperature	Input × Grad	$\mathbf{-0.141}$ $[\mathbf{-0.168}, \mathbf{-0.114}]$	$\mathbf{3.1\times10^{-21}}$
Qwen2.5 1.5B	IWSLT	Temperature	IG	-0.025 $[-0.052,\ \ 0.002]$	0.12
Qwen2.5 3B	LAMBADA	Temperature	Input × Grad	$\mathbf{-0.162}$ $[\mathbf{-0.187}, \mathbf{-0.136}]$	$\mathbf{1.5\times10^{-32}}$
Qwen2.5 3B	IWSLT	Temperature	IG	$\mathbf{-0.081}$ $[\mathbf{-0.118}, \mathbf{-0.044}]$	$\mathbf{9.3\times10^{-6}}$
Gemma-3 1B	LAMBADA	Fisher	Input × Grad	$\mathbf{-0.109}$ $[\mathbf{-0.126}, \mathbf{-0.091}]$	$\mathbf{3.7\times10^{-30}}$
Gemma-3 1B	IWSLT	Fisher	Input × Grad	$\mathbf{-0.037}$ $[\mathbf{-0.051}, \mathbf{-0.023}]$	$\mathbf{1.1\times10^{-4}}$
Gemma-3 4B	LAMBADA	Fisher	Input × Grad	$\mathbf{-0.110}$ $[\mathbf{-0.127}, \mathbf{-0.094}]$	$\mathbf{8.0\times10^{-32}}$
Gemma-3 4B	IWSLT	Fisher	IG	$\mathbf{0.145}$ $[\ \mathbf{0.112},\ \ \mathbf{0.178}]$	$\mathbf{9.4\times10^{-18}}$

A.4 Additional expamples

To demonstrate the prowess of Jacobian Scopes on analyzing long-range interaction across numerous tokens, we provide additional examples on a larger model, LLaMA-3.2 3B.

Semantic Scope: system-prompt instructions. Figure 6 shows Semantic Scope applied to a system-prompt-style instruction on LLaMA-3.2 1B, illustrating how the model links output tokens to semantically relevant input tokens.

Fisher Scope: word-level translation. We prompt the LLM to perform translations by providing it with the following structure: "`name of original language`": `[original text]`. "`name of target language`": Figure 7 shows a simpler example complementing Figure 3: here every predicted token is unambiguously attributed to its direct lexical counterpart in the source language, illustrating the word-level translation regime.

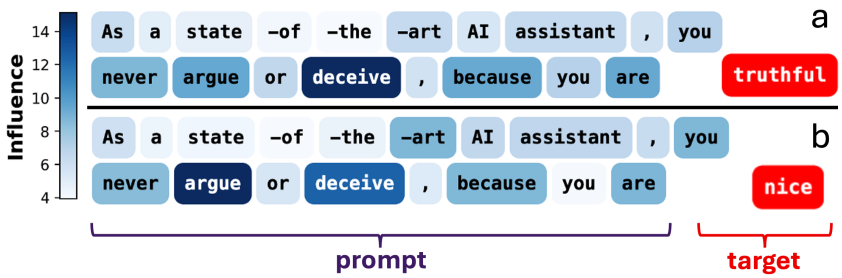


Figure 6: Semantic Scope visualizes how LLaMA-3.2 understands system-prompt-style instructions. The logit assigned to the token "truthful" is most prominently attributed to the input token "deceive", and "nice" to "argue".

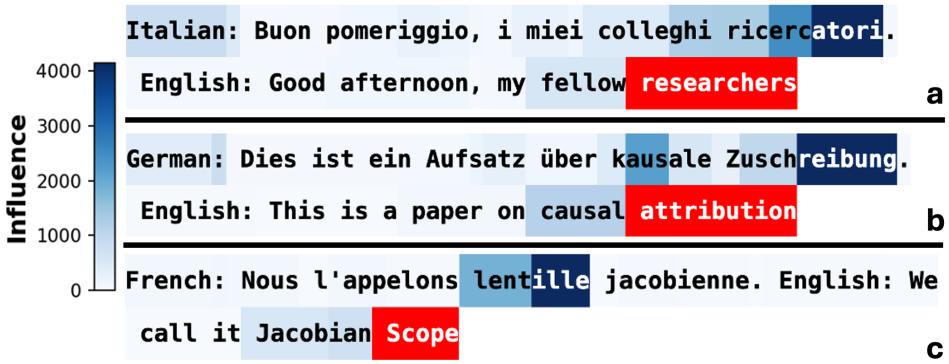


Figure 7: Fisher Scope reveals that translated words, highlighted in red, are primarily influenced by their lexical counterparts in the source language.

Semantic Scope: Solving a crime mystery. To further showcase the capability of Jacobian Scopes in analyzing LLM reasoning over extended contexts, we constructed a detective micro-mystery, excerpted below:

> Since the victim, a wealthy aristocrat living in the English countryside, died from a fatal wound, the detective narrowed down the three possible murder weapons: a cleaver, a shovel, and a screwdriver. He further deduced that the killer is the ...

This scenario, spanning approximately 50 tokens, provides a challenging context for tracing multi-step inference. Applying Semantic Scope, we found that LLaMA-3.2 correctly links each suspect with the relevant evidence, as visualized in Figure 8.

A.5 Dynamical systems: definitions and additional experiments

This section provides the mathematical definition for all dynamical systems investigated. All attributions in this section are performed via Temperature Scope as detailed in section 4.3.

Logistic map. The logistic map is a simple discrete-time dynamical system that exhibits chaotic behavior [40]. It is governed by an iterative equation:

$$x_{t+1} = f(x_t) = r x_t (1 - x_t), \quad x_0 \in (0, 1),$$

where $r \in [1, 4)$ is a hyper-parameter. In our experiment we set $r = 3.8$, which is deep inside the chaotic regime of the system. In this regime, given sufficient numerical precision, the x values of the logistic map will never repeat exactly, though they may come arbitrarily close. This behavior, known as *aperiodicity* [40], results in the recurrence of similar patterns without ever producing an identical sequence. As illustrated in Figure 9, LLaMA-3.2 extrapolates the logistic map by attending to such recurring motifs in the sequence's history — adopting a similar strategy as it uses for in-context learning the Lorenz system (Figure 4).

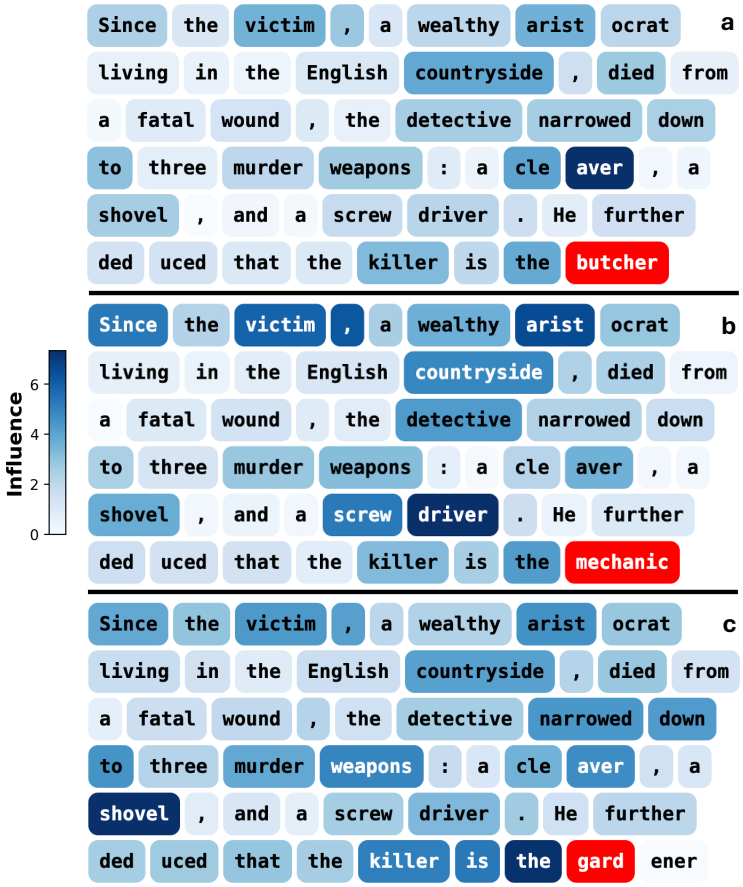


Figure 8: **Semantic Scope attribution for detective micro-fiction.** The attribution graph illustrates how LLaMA-3.2 3B connects suspects to their associated clues when reasoning about potential perpetrators and weapons in a novel-like setting. Attributions clarify the model's process for linking evidence to conclusions across a long narrative context.

Lorenz system. The Lorenz system [26] is a classic 3D dynamical model from atmospheric science, defined by:

$$\begin{aligned}\dot{x}(t) &= \sigma(y - x),\\ \dot{y}(t) &= x(\rho - z) - y,\\ \dot{z}(t) &= xy - \beta z,\end{aligned}$$

where $\sigma = 10$, $\rho = 28$, and $\beta = 8/3$ produce chaotic behavior. We simulate the system with a first-order explicit scheme, varying the initial x while fixing y and z. The system's chaotic nature ensures that even tiny differences in initial conditions lead to rapidly diverging trajectories. The LLM is then prompted with the tokenized x series. The resulting x values show characteristic *aperiodic* oscillations between two lobes of the well-known butterfly attractor.

Figures 4 and 10 illustrate the model's behavior under two different initial conditions. In Figure 4 (a), the cutoff occurs at a peak in the x value, while in Figure 10, the cutoff coincides with the x value transitioning between attractor lobes. In both scenarios, the LLM attends to similar patterns surrounding the cutoff region.

Lorenz system with drift. To verify whether LLMs overlook the part of in-context data that is a statistical outlier, we add a uniform drift term in the same Lorenz system shown in Figure 4 (a). As such a system evolves, the later part of the contexts will eventually wander *out of distribution* compared to the earlier part. The resulting attribution pattern in Figure 11 indeed shows diminished influence score at the earlier regions in the context window.

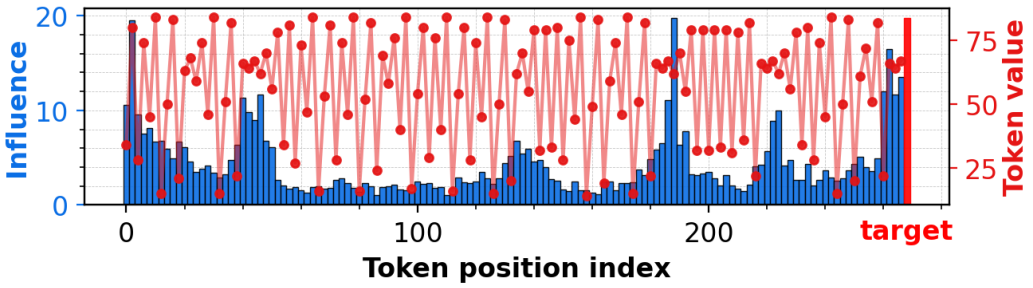


Figure 9: Temperature Scope shows LLaMA-3.2 attending to similar motifs in history when extrapolating a logistic map trajectory.

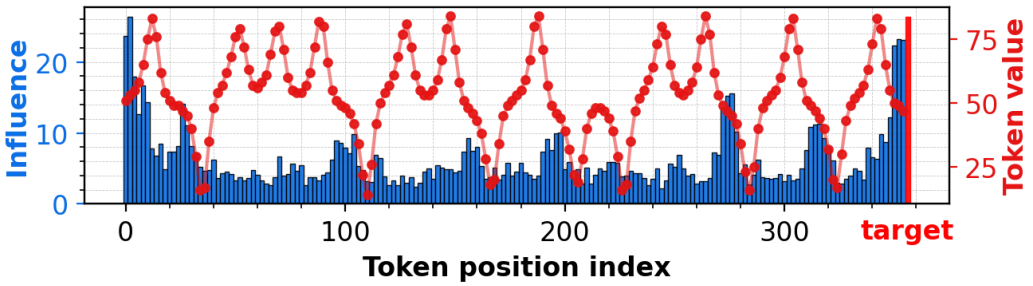


Figure 10: Temperature Scope attribution graph of LLaMA-3.2 extrapolating a partially observed Lorenz system, with different initial condition than Figure 4 (a).

Additional examples on Brownian motion. Brownian motion [11], or Wiener process, is a classic model for random movement, defined by the SDE:

$$dX_t = \mu\, dt + \sigma\, dW_t, \tag{14}$$

where μ is drift and σ is diffusion. For $\mu = 0$, $\sigma = 1$, this gives standard Brownian motion. Figure 12 displays three additional Temperature Scope attribution plots for LLaMA-3.2 extrapolating Brownian motion sequences, each generated using a different random seed. As discussed in section 4.3 and [24], the LLM assigns higher influence scores to tokens towards the end of the context window — likely because the earlier portions become increasingly out-of-distribution as the sequence evolves.

LLM attends to spurious patterns in Brownian motion. Notably, while the overall influence trend increases with context length — as seen in Figure 12 — there are many abrupt changes and irregularities in the influence score. These anomalies can be explained by random fluctuations: occasionally, patterns or trends may reappear in Brownian motion purely by statistical chance. For instance, in panel (b) of Figure 4, there is a sudden drop in x-values around token positions 100, which is "echoed" by a similar drop around the cutoff point at 360. Much like an amateur stock trader who perceives random fluctuations in stock movements as meaningful signals, the LLM incorrectly interprets these coincidental recurrences as informative patterns.

Together, these results illustrate that while LLMs are capable of emergent ICL behaviors such as pattern matching, they may also exhibit human-like tendencies toward superstition [14], and perceive spurious patterns in random numerical sequences.

A.6 Temperature Scope is optimal for numerical data

As argued in the main text, Temperature Scope is more appropriate for time-series forecasting tasks, where the object of interest is the full next-step distribution rather than any individual token. We illustrate this distinction with a concrete example in Figure 13. Conditioned on the first T inputs, the model's prediction for the $(T+1)$st state exhibits a bell-shaped distribution over approximately 20 numerical tokens, consistent with prior observations in [24]. The ground-truth value corresponds to only one of many plausible candidates within this distribution.

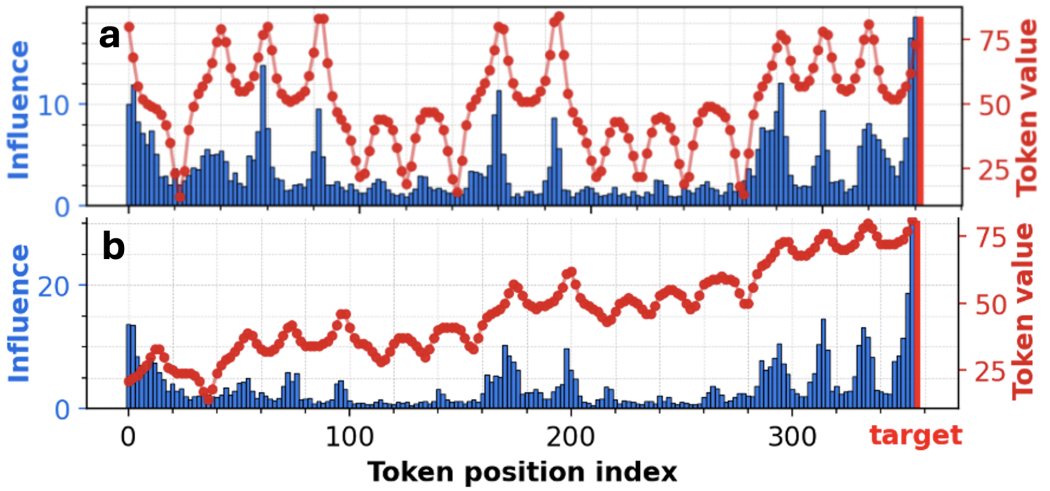


Figure 11: Temperature Scope attribution graph of LLaMA-3.2 extrapolating (a) a partially observed Lorenz system, and (b) the same Lorenz system with uniform drift.

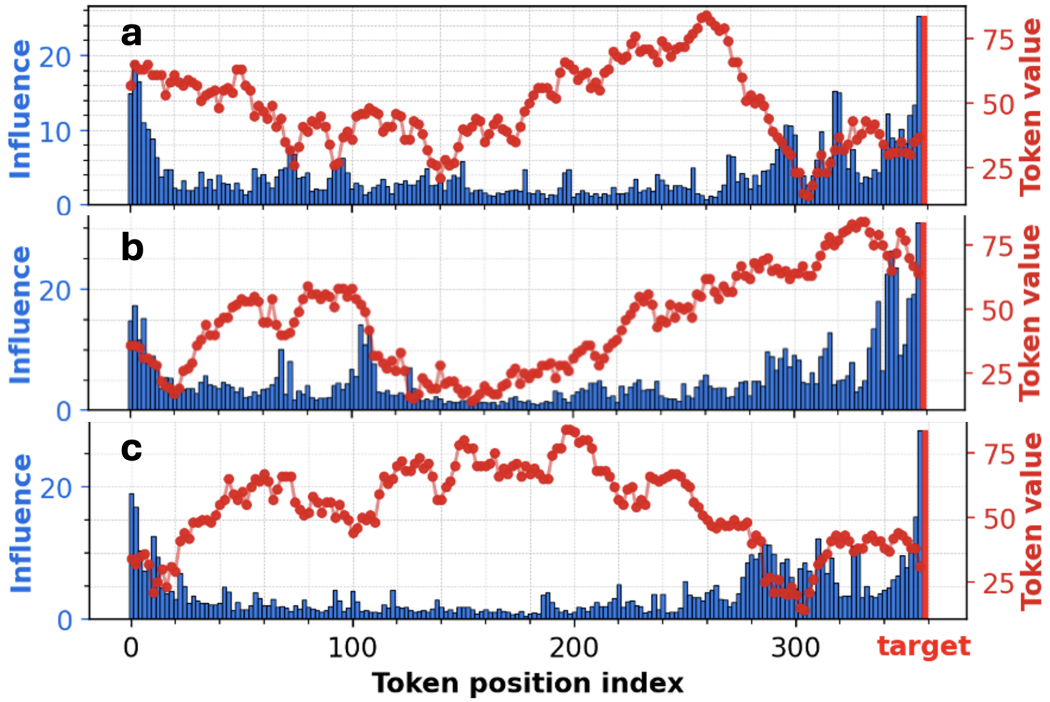


Figure 12: Additional Temperature Scope attributions on LLaMA-3.2 extrapolating Brownian motions in-context.

Naively applying Semantic Scope with the ground-truth token (e.g. “84”) as the target yields misleading attributions: the resulting influence map highlights only the inputs that increase the logit of “84”, while ignoring the remaining tokens that collectively constitute the bell-shaped predictive distribution. As a result, the attribution fails to capture the model’s reasoning at the distributional level.

Fisher Scope also exhibits attenuated attribution signals at early context positions. A heuristic explanation is as follows. Fisher Scope quantifies how input perturbations affect the predictive distribution over the *entire vocabulary*, including both numerical and non-numerical tokens. However, the underlying dynamical system is defined only over numerical tokens. Early numerical inputs

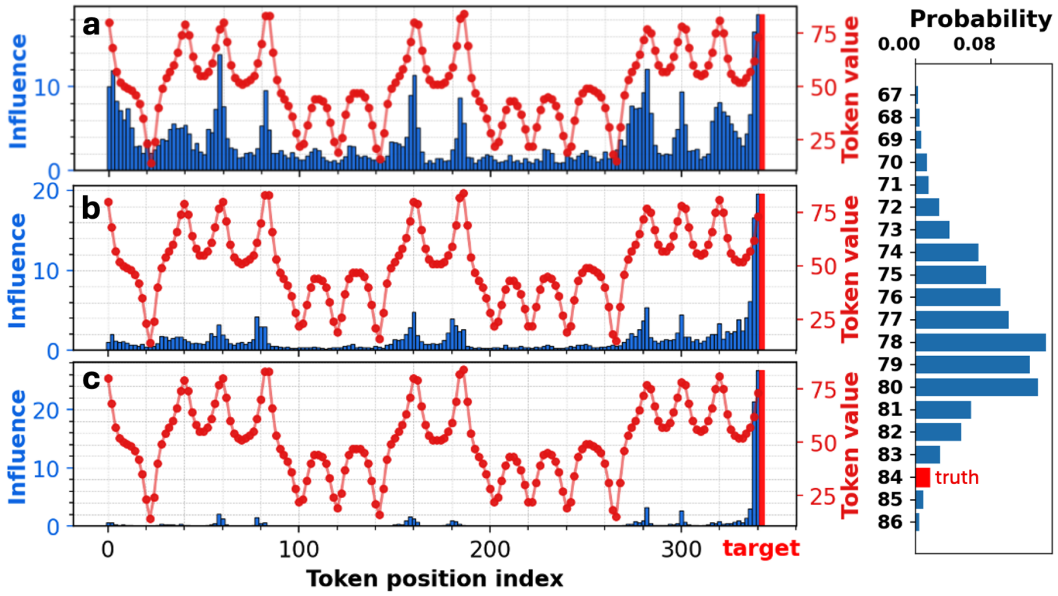


Figure 13: Left: (a) Temperature Scope (b) Semantic Scope and (c) Fisher Scope attributions for the same Lorenz system. Right: The predicted probability mass over the top 20 candidate tokens at the target position, ordered by their numerical values. The ground-truth token (shown in red) is used as the target for Semantic Scope.

therefore inform the implicit transition rule learned by the LLM [24] and can significantly influence future numerical predictions. Yet these inputs carry little information about non-numerical tokens, whose probabilities tend to default to patterns determined by near-term context. Because Fisher Scope aggregates sensitivity across the entire vocabulary, this mismatch dilutes the attribution signal originating from early numerical tokens.

Temperature Scope, by contrast, is dominated by the most probable tokens, which in this setting are numerical tokens. It therefore focuses attribution on the portion of the distribution that is relevant to the forecasting task, effectively bypassing the out-of-support vocabulary. This makes Temperature Scope better aligned with forecasting scenarios in which uncertainty and variance are intrinsic to the prediction.

A.7 Effective temperature as variance

We prove proposition 1 stated in section 4.3.

Proof. ($\Leftarrow$) Suppose $\hat{z}(v) = -b(v-\mu)^2 + c$. Then

$$e^{\beta\,\hat{z}(v)} = e^{\beta c} \cdot e^{-\beta b(v-\mu)^2},$$

which is a Gaussian kernel with mean μ and variance $\sigma^2 = 1/(2\beta b)$.

($\Rightarrow$) Suppose $p(v)$ is Gaussian with mean μ and variance σ^2, so that $e^{\beta\,\hat{z}(v)} \propto e^{-(v-\mu)^2/(2\sigma^2)}$. Taking logarithms gives

$$\hat{z}(v) = -\frac{1}{2\beta\sigma^2}(v-\mu)^2 + \text{const},$$

which is quadratic with $b = 1/(2\beta\sigma^2)$, recovering $\sigma^2 = 1/(2\beta b)$. □

Proposition 1 provides a direct justification for using Temperature Scope in time-series forecasting. Since the predictive distribution over numerical tokens is approximately Gaussian — as empirically observed in Figures 4 and 13 — β_{eff} is the quantity that controls the width of that distribution. Attributing changes in β_{eff} therefore directly identifies which input tokens govern predictive uncertainty, which is precisely what Temperature Scope computes.

Entropy derivative. For completeness we record the short derivation of eq. (9). With $p(i;\beta) = e^{\beta\hat{z}(i)}/Z(\beta)$ and $Z = \sum_j e^{\beta\hat{z}(j)}$,

$$H(\beta) = -\sum_i p(i)\big(\beta\hat{z}(i) - \log Z\big) = -\beta\,\mathbb{E}_p[\hat{z}] + \log Z,$$

$$\frac{dH}{d\beta} = -\mathbb{E}_p[\hat{z}] - \beta\,\mathrm{Cov}_p(\hat{z},\hat{z}) + \mathbb{E}_p[\hat{z}] = -\beta\,V(\beta),$$

where we used $\partial_\beta \log Z = \mathbb{E}_p[\hat{z}]$ and $\partial_\beta \mathbb{E}_p[\hat{z}] = V(\beta)$.

A.8 Path-Integrated Semantic Scope

Integrated Gradients (IG) [41] is a widely used gradient-based attribution method, with successful applications across computer vision, molecular modeling, and language models. By integrating gradients along a continuous path between a baseline input and the full input, IG aims to mitigate attribution noise arising from non-smoothness ("kinks") in deep networks. In this section, we compare the original Semantic Scope with its path-integrated analogue and show that, in the LLM setting, IG introduces distortions due to the out-of-distribution portion of the integration path.

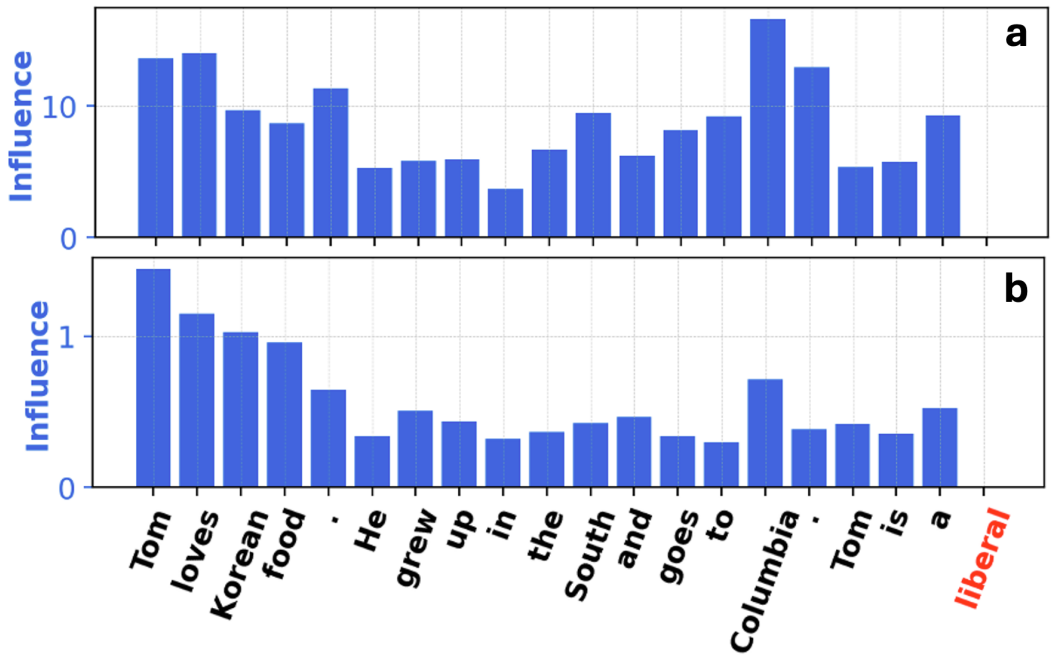


Figure 14: Causal attribution using (a) Semantic Scope and (b) its path-integrated variant based on Integrated Gradients.

To apply Integrated Gradients, one first defines a linear path parameterized by $\alpha \in [0,1]$ that connects two matrices $\boldsymbol{X} = \boldsymbol{X}_{1:T}$ and $\boldsymbol{X}' = 0$ (the full input and the null baseline):

$$\tilde{\boldsymbol{X}}(\alpha) = \boldsymbol{X}' + \alpha(\boldsymbol{X} - \boldsymbol{X}').$$

One then selects a scalar output $z \in \mathbb{R}$ as the explanandum. In our setting, $\boldsymbol{X} \in \mathbb{R}^{d_{\text{model}} \times T}$ denotes the full input sequence of token embeddings, and z is the logit associated with a target vocabulary item (e.g., "liberal" or "conservative") [2].

Consistent with Sundararajan et al. [41], the Integrated Gradient attribution is defined as

$$\mathrm{IG}(\boldsymbol{X}) := (\boldsymbol{X} - \boldsymbol{X}') \odot \int_0^1 \nabla z\Big(\tilde{\boldsymbol{X}}(\alpha)\Big)\, d\alpha, \tag{15}$$

[2] In the original IG formulation [41], $\boldsymbol{X}$ represents image pixels and z corresponds to the logit or log-probability of a predicted class (e.g., "truck" or "boat").

where $\odot$ denotes element-wise multiplication. This yields an attribution matrix $\mathrm{IG}(\boldsymbol{X}) \in \mathbb{R}^{d_{\text{model}} \times T}$. In direct analogy with eq. (1), we define the path-integrated influence score for token t as the l^2-norm of a vector, namely

$$\mathrm{Influence}_t^{\mathrm{IG}} := \|\mathrm{IG}(\boldsymbol{X})_t\|_2 \,, \tag{16}$$

where $\mathrm{IG}(\boldsymbol{X})_t$ denotes the tth column, corresponding to the integrated gradient of the input embedding $\boldsymbol{x}_t$.

Figure 14 compares the resulting IG attribution with the original Semantic Scope on the example discussed in section 4.1. Relative to Semantic Scope, the IG-based method assigns disproportionately large influence to early-context tokens, thereby obscuring the semantically decisive input token "Columbia."

To diagnose the origin of this behavior, we visualize the integrand $\nabla z(\tilde{\boldsymbol{X}}(\alpha))$ at several values of α in Figure 15. Specifically, for each α we plot the token-wise ℓ_2 norm of the gradient, $\|\nabla_{\boldsymbol{x}_t} z(\tilde{\boldsymbol{X}}(\alpha))\|_2$, consistent with the influence definition in eq. (16). In practice, the integral in eq. (15) is approximated by a discrete sum over 100 uniformly spaced values of α; the figure shows a subset of these values.

As a sanity check, we observe that for $\alpha \to 1$, the integrand converges to the Semantic Scope attribution: $\nabla z(\tilde{\boldsymbol{X}}(1)) = \nabla z(\boldsymbol{X})$. In contrast, at small α (e.g., $\alpha \approx 0.01$), the input sequence is nearly indistinguishable from the null input. In this regime, the model operates far outside the data manifold, and circuit-level artifacts dominate. In particular, the "attention sink" phenomenon [48] causes attribution to be overwhelmingly concentrated on early token positions, with influence scores decaying approximately exponentially over context length.

Crucially, the gradient magnitudes in this low-α regime are also large, causing these out-of-distribution effects to dominate the path integral in eq. (15). At intermediate values of α, the attribution is often highest for function words such as "and," which are not semantically relevant to the prediction. Taken together, these observations indicate that the IG path spends substantial measure in regions of representation space that are both out-of-distribution and attributionally pathological.

In summary, while Integrated Gradients can be effective when the interpolation path remains in-distribution, it performs poorly for LLMs. The null-sentence regime introduces severe distortions — most notably attention sink effects — that dominate the attribution signal. Moreover, this degradation comes at a substantial computational cost: in practice, IG requires approximately $100\times$ more computation than Semantic Scope due to the discrete approximation of the path integral. As a result, path integration both worsens attribution quality and significantly increases computational overhead in the settings explored in this work.

A.9 Geometric interpretation of influence under ε -norm perturbations

We provide a precise justification for the geometric interpretation of the influence score defined in eq. (1). Recall that $\boldsymbol{J}_t := \partial \boldsymbol{y} / \partial \boldsymbol{x}_t \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ denotes the Jacobian of the leading hidden state $\boldsymbol{y}$ with respect to the input embedding $\boldsymbol{x}_t$, and let $\boldsymbol{v} \in \mathbb{R}^{d_{\text{model}}}$ denote a fixed direction in output space.

Under a first-order Taylor expansion, a small perturbation $\boldsymbol{\delta x_t}$ to the input embedding induces a change in the output

$$\boldsymbol{\delta y} \approx \boldsymbol{J}_t \boldsymbol{\delta x}_t.$$

Projecting this change onto the direction $\boldsymbol{v}$ yields the scalar response

$$\boldsymbol{v}^\intercal \boldsymbol{\delta y} \;\approx\; \boldsymbol{v}^\intercal \boldsymbol{J}_t \boldsymbol{\delta x_t}.$$

We are interested in the maximum possible change along $\boldsymbol{v}$ induced by any perturbation of bounded norm, i.e.,

$$\max_{\|\boldsymbol{\delta x}_t\|_2 \le \varepsilon} \boldsymbol{v}^\intercal \boldsymbol{J}_t \boldsymbol{\delta x}_t.$$

By the Cauchy–Schwarz inequality, this quantity is upper-bounded by

$$\boldsymbol{v}^\intercal \boldsymbol{J}_t \boldsymbol{\delta x}_t \;\le\; \|\boldsymbol{v}^\intercal \boldsymbol{J}_t\|_2 \, \|\boldsymbol{\delta x}_t\|_2 \;\le\; \varepsilon \, \|\boldsymbol{v}^\intercal \boldsymbol{J}_t\|_2,$$

with equality achieved when $\boldsymbol{\delta x}_t$ is aligned with $(\boldsymbol{v}^\intercal \boldsymbol{J}_t)^\intercal$. Therefore,

$$\max_{\|\boldsymbol{\delta x}_t\|_2 \le \varepsilon} \boldsymbol{v}^\intercal \boldsymbol{J}_t \boldsymbol{\delta x}_t = \varepsilon \, \|\boldsymbol{v}^\intercal \boldsymbol{J}_t\|_2.$$

This shows that $\|\boldsymbol{v}^\intercal \boldsymbol{J}_t\|_2$ quantifies the largest possible first-order magnification, along the direction $\boldsymbol{v}$, that can be induced by an ε-norm perturbation to the input token embedding $\boldsymbol{x}_t$, up to the scale factor ε. This motivates the definition of the influence score in eq. (1).

A.10 Second-order KL expansion and Fisher pullback

Let $\boldsymbol{p}(\cdot \mid \boldsymbol{x}) \in \mathbb{R}^{|\mathcal{V}|}$ denote a smooth family of categorical distributions parameterized by an input $\boldsymbol{x}_t$. For a small perturbation $\boldsymbol{\delta x}_t$, consider the KL divergence

$$\begin{aligned}&\mathrm{KL}(\boldsymbol{p}(\cdot \mid \boldsymbol{x}_t) \,\|\, \boldsymbol{p}(\cdot \mid \boldsymbol{x}_t + \boldsymbol{\delta x}_t))\\ &= \sum_i \boldsymbol{p}(i \mid \boldsymbol{x}_t)\Big(\log \boldsymbol{p}(i \mid \boldsymbol{x}_t) - \log \boldsymbol{p}(i \mid \boldsymbol{x}_t + \boldsymbol{\delta x}_t)\Big).\end{aligned}$$

Under standard regularity conditions, the first-order term vanishes and the KL divergence admits the second-order expansion

$$\mathrm{KL}(\boldsymbol{p}(\cdot \mid \boldsymbol{x}_t) \,\|\, \boldsymbol{p}(\cdot \mid \boldsymbol{x}_t + \boldsymbol{\delta x}_t)) = \frac{1}{2}\, \boldsymbol{\delta x}_t^\intercal \boldsymbol{F}(\boldsymbol{x}_t) \boldsymbol{\delta x}_t + o\big(\|\boldsymbol{\delta x}_t\|^2\big)\,,$$

where the Fisher information matrix is given by [3]

$$\boldsymbol{F}_t = \boldsymbol{F}(\boldsymbol{x}_t) := \mathbb{E}_{\boldsymbol{p}(\cdot|\boldsymbol{x}_t)}[(\nabla_{\boldsymbol{x}_t} \log \boldsymbol{p})(\nabla_{\boldsymbol{x}_t} \log \boldsymbol{p})^\intercal]\,.$$

We now derive the close-form expression for $\boldsymbol{F}_t$. For a categorical distribution $\boldsymbol{p} = \mathrm{softmax}(\boldsymbol{z})$, parameterized by logits $\boldsymbol{z} \in \mathbb{R}^{|\mathcal{V}|}$, its Fisher information with respect to $\boldsymbol{z}$ takes the closed form [6]

$$\boldsymbol{F}(\boldsymbol{z}) = \mathrm{Diag}(\boldsymbol{p}) - \boldsymbol{p}\boldsymbol{p}^\intercal.$$

If the logits are given by $\boldsymbol{z} = \boldsymbol{W}\boldsymbol{y}$, then the Fisher information with respect to the hidden state $\boldsymbol{y}$ is

$$\boldsymbol{F} := \boldsymbol{F}(\boldsymbol{y}) = \boldsymbol{W}^\intercal\big(\mathrm{Diag}(\boldsymbol{p}) - \boldsymbol{p}\boldsymbol{p}^\intercal\big)\boldsymbol{W}.$$

Finally, to assess how an input token $\boldsymbol{x}_t$ influences the model's prediction, we pull back [5] the Fisher information matrix through the Jacobian $\boldsymbol{J}_t$ to the embedding space of token t:

$$\boldsymbol{F}_t = \boldsymbol{J}_t^\intercal \boldsymbol{F} \boldsymbol{J}_t \tag{17}$$

A.11 Fisher Scope as a low-rank approximation of total information

The Fisher Scope influence score in eq. (6) is based on the principal Fisher direction $\boldsymbol{u}_1$, the single direction in output space most sensitive to changes in the predictive distribution. Here we show that this can be understood as the leading term of a more general quantity, which aggregates distributional sensitivity across all directions.

Total information as expected KL divergence.

Consider an isotropic unit-norm perturbation $\boldsymbol{\delta x}_t = \varepsilon \boldsymbol{u}$, where $\varepsilon > 0$ is small and $\boldsymbol{u} \in \mathbb{R}^d$ is uniform on the unit sphere. Taking expectation and using $\mathbb{E}_{\boldsymbol{u}}[\boldsymbol{u}\boldsymbol{u}^\intercal] = \frac{1}{d}\boldsymbol{I}$ yields

$$\begin{aligned}&\mathbb{E}_{\boldsymbol{u}}[\mathrm{KL}(\boldsymbol{p}(\cdot \mid \boldsymbol{x}_t) \,\|\, \boldsymbol{p}(\cdot \mid \boldsymbol{x}_t + \varepsilon \boldsymbol{u}))]\\ &= \frac{1}{2}\,\varepsilon^2\, \mathbb{E}_{\boldsymbol{u}}[\boldsymbol{u}^\intercal \boldsymbol{F}_t\, \boldsymbol{u}] + o(\varepsilon^2)\\ &= \frac{1}{2}\,\varepsilon^2\, \mathrm{tr}\Big(\boldsymbol{F}_t\, \mathbb{E}_{\boldsymbol{u}}[\boldsymbol{u}\boldsymbol{u}^\intercal]\Big) + o(\varepsilon^2)\\ &= \frac{1}{2}\,\varepsilon^2\, \frac{1}{d}\,\mathrm{tr}(\boldsymbol{F}_t) + o(\varepsilon^2).\end{aligned} \tag{18}$$

Thus, up to the universal factor $\varepsilon^2/(2d)$, the trace $\mathrm{tr}(\boldsymbol{F}_t)$ governs the expected KL divergence induced by an isotropic perturbation of the input token $\boldsymbol{x}_t$. We therefore refer to $\mathrm{tr}(\boldsymbol{F}_t)$ as the *total information* at position t.

Fisher Scope as rank-1 approximation. Since $\boldsymbol{F}$ admits the eigen-decomposition $\boldsymbol{F} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^\intercal$, the total information can be written exactly as

$$\mathrm{tr}(\boldsymbol{F}_t) = \sum_{i=1}^{d_{\mathrm{model}}} \lambda_i \|\boldsymbol{u}_i^\intercal \boldsymbol{J}_t\|_2^2. \tag{19}$$

In practice, $\boldsymbol{F}$ admits a good low-rank approximation $\boldsymbol{F} \approx \boldsymbol{U}_r \boldsymbol{\Lambda}_r \boldsymbol{U}_r^\intercal$, where $\boldsymbol{U}_r \in \mathbb{R}^{d_{\text{model}} \times r}$ contains the top r eigenvectors and $\boldsymbol{\Lambda}_r$ the corresponding eigenvalues. This yields the approximation

$$\operatorname{tr}(\boldsymbol{F}_t) \approx \sum_{i=1}^{r} \lambda_i \|\boldsymbol{u}_i^\intercal \boldsymbol{J}_t\|_2^2, \tag{20}$$

where each term $\|\boldsymbol{u}_i^\intercal \boldsymbol{J}_t\|_2$ requires only a single backward pass, analogous to eq. (2). The full total information thus requires r backward passes, compared to d_{model} passes needed to form $\boldsymbol{J}_t$ explicitly.

The Fisher Scope influence score in eq. (6) corresponds to the $r = 1$ case of eq. (20), retaining only the leading term $\lambda_1 \|\boldsymbol{u}_1^\intercal \boldsymbol{J}_t\|_2^2$ and dropping the scalar λ_1 since it is constant across token positions t.

Low-rankness of $\boldsymbol{F}$. Figure 16 provides three complementary pieces of evidence that the $r = 1$ approximation is justified in practice. The top panel shows the eigenvalue spectrum of $\boldsymbol{F}$ across 10 LAMBADA prompts on a log-log scale: a consistent power-law decay spanning 6 orders of magnitude, indicating that the leading eigenvalue dominates all subsequent ones.

The middle panel confirms this quantitatively via the cumulative explained variance $\sum_{i=1}^{r} \lambda_i / \operatorname{tr}(\boldsymbol{F})$, averaged over 1000 LAMBADA passages: at $r = 1$ the mean explained fraction is already $\approx 37\%$, rising to $\approx 65\%$ by $r = 10$ and approaching 1 only beyond $r \approx 1000$.

The bottom panel provides direct empirical justification, extending the LLaMA-3.2 1B results in table 1: rank-1 Fisher Scope ($r = 1$) already outperforms Input $\times$ Gradient and Semantic Scope on LAMBADA, and matches Temperature Scope. Increasing the rank to $r \in \{4, 16, 64, 256\}$ yields only marginal further gains in AOPC, while multiplying the computational cost by r.

Together, these results justify both the privileged status of $\boldsymbol{u}_1$ and the choice of $r = 1$ in Fisher Scope.

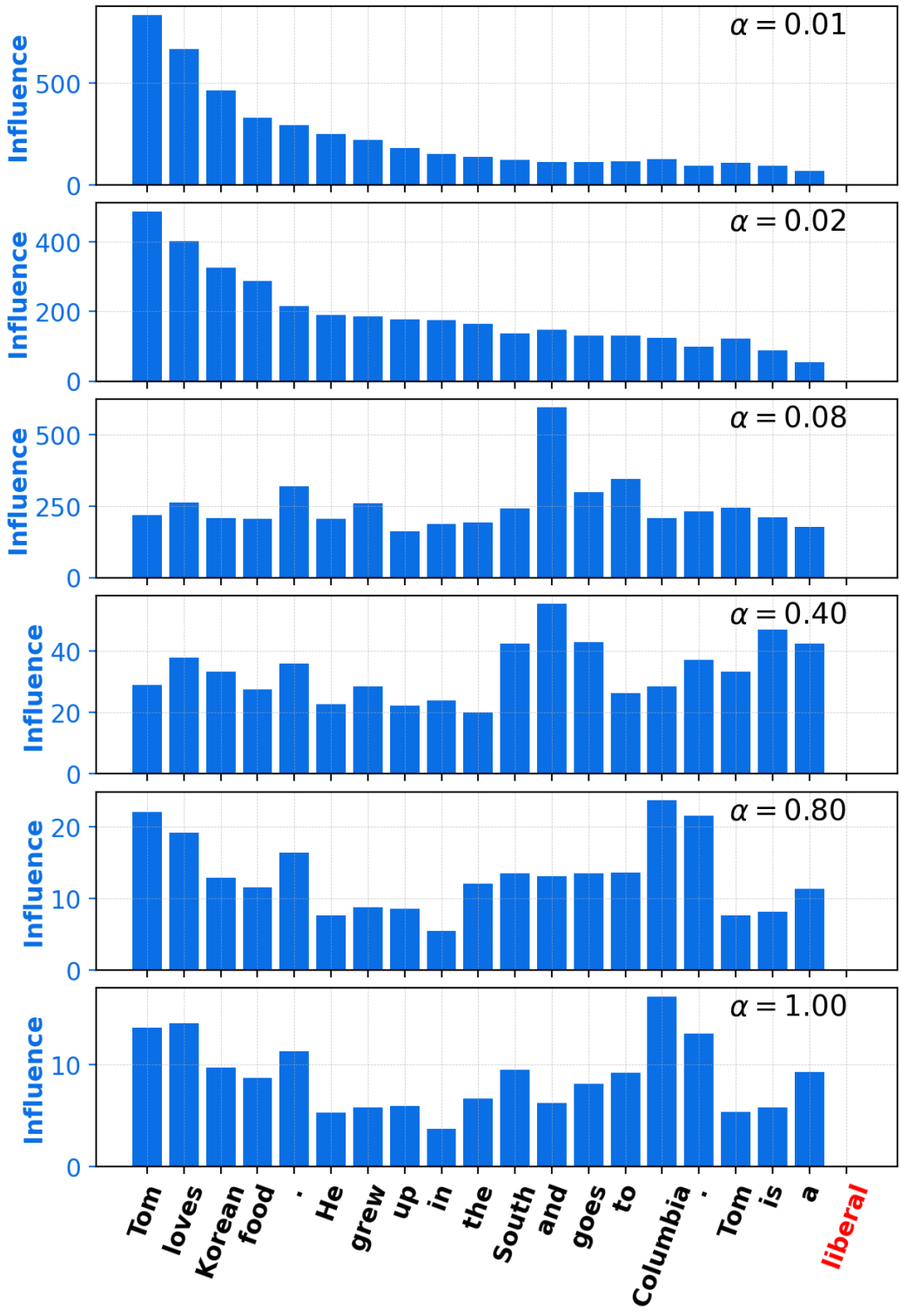


Figure 15: Token-wise ℓ_2 norm of the Integrated Gradient integrand $\nabla z(\tilde{\boldsymbol{X}}(\alpha))$ along the interpolation path $\alpha \in [0, 1]$.

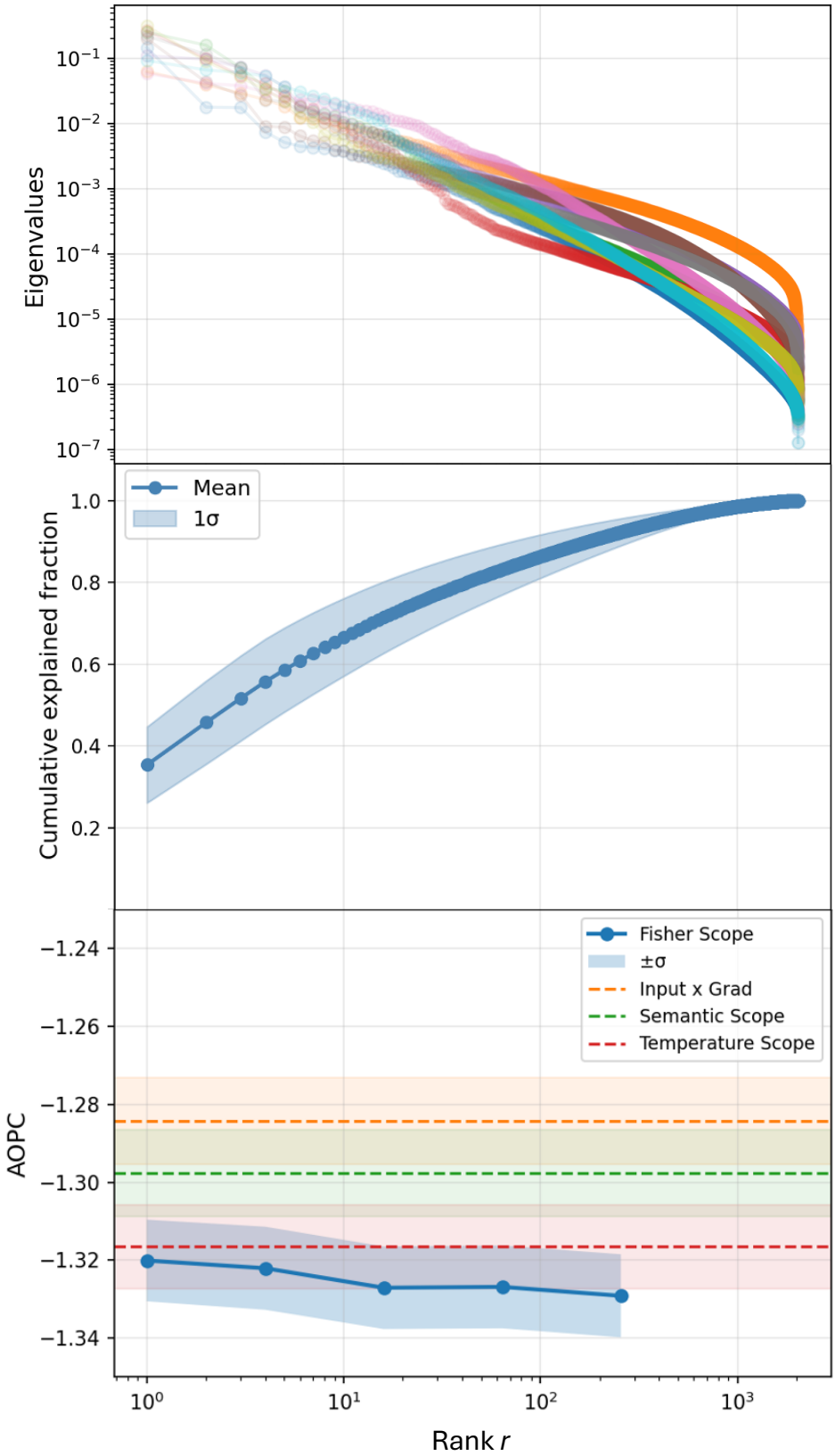


Figure 16: Justification for the rank-1 approximation used in Fisher Scope, evaluated on LLaMA-3.2 1B and LAMBADA. **Top:** Eigenvalue spectrum of $\boldsymbol{F}$ across 10 prompts on a log-log scale, exhibiting a consistent power-law decay spanning 6 orders of magnitude. **Middle:** Mean cumulative explained variance $\sum_{i=1}^{r} \lambda_i / \mathrm{tr}(\boldsymbol{F})$ averaged over 1000 passages (shaded band: $\pm 1\sigma$); $\boldsymbol{u}_1$ alone accounts for $\approx$37% of total sensitivity (1σ: 25–50%). **Bottom:** AOPC as a function of rank $r \in \{1, 4, 16, 64, 256\}$, with baselines shown as dashed lines. Rank-1 Fisher Scope already outperforms Input $\times$ Gradient and Semantic Scope; increasing r yields only marginal improvement at r-fold computational cost.

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